The attached tables are chemical 'results' and 'criteria'.
The criteria join to the results using the 'Chem Name' column in both tables.
Write me an Excel formula that compares the 'Result' column in the results table against the HIL-A criteria values in the 'Action Level' column in the criteria table.
Convert the '<' results to half when comparing values to criteria

```
Location Code,Depth,Sampled Date,Chem Name,Prefix,Result,Unit
BH01,3 - 3.1,15/11/2023,Arsenic,,45,mg/kg
BH01,3 - 3.1,15/11/2023,Cadmium,,24,mg/kg
BH01,3 - 3.1,15/11/2023,Chromium,,615,mg/kg
BH01,3 - 3.1,15/11/2023,Copper,,253,mg/kg
BH01,3 - 3.1,15/11/2023,Lead,,1800,mg/kg
BH01,3 - 3.1,15/11/2023,Mercury,,0.1,mg/kg
BH01,3 - 3.1,15/11/2023,Nickel,,694,mg/kg
BH01,3 - 3.1,15/11/2023,Zinc,,2340,mg/kg
BH01,4.8 - 4.9,15/11/2023,Arsenic,,26,mg/kg
BH01,4.8 - 4.9,15/11/2023,Cadmium,,5,mg/kg
BH01,4.8 - 4.9,15/11/2023,Chromium,,50,mg/kg
BH01,4.8 - 4.9,15/11/2023,Copper,,2150,mg/kg
BH01,4.8 - 4.9,15/11/2023,Lead,,837,mg/kg
BH01,4.8 - 4.9,15/11/2023,Mercury,,0.2,mg/kg
BH01,4.8 - 4.9,15/11/2023,Nickel,,88,mg/kg
BH01,4.8 - 4.9,15/11/2023,Zinc,,5310,mg/kg
BH03,0.5 - 0.6,15/11/2023,Arsenic,,11,mg/kg
BH03,0.5 - 0.6,15/11/2023,Cadmium,<,1,mg/kg
BH03,0.5 - 0.6,15/11/2023,Chromium,,42,mg/kg
BH03,0.5 - 0.6,15/11/2023,Copper,,23,mg/kg
BH03,0.5 - 0.6,15/11/2023,Lead,,22,mg/kg
BH03,0.5 - 0.6,15/11/2023,Mercury,<,0.1,mg/kg
BH03,0.5 - 0.6,15/11/2023,Nickel,,37,mg/kg
BH03,0.5 - 0.6,15/11/2023,Zinc,,72,mg/kg
BH03,2 - 2.1,15/11/2023,Arsenic,,6,mg/kg
BH03,2 - 2.1,15/11/2023,Cadmium,<,1,mg/kg
BH03,2 - 2.1,15/11/2023,Chromium,,28,mg/kg
BH03,2 - 2.1,15/11/2023,Copper,,49,mg/kg
BH03,2 - 2.1,15/11/2023,Lead,,184,mg/kg
BH03,2 - 2.1,15/11/2023,Mercury,<,0.1,mg/kg
BH03,2 - 2.1,15/11/2023,Nickel,,26,mg/kg
BH03,2 - 2.1,15/11/2023,Zinc,,225,mg/kg
BH05,1.8 - 1.9,15/11/2023,Arsenic,,5,mg/kg
BH05,1.8 - 1.9,15/11/2023,Cadmium,<,1,mg/kg
BH05,1.8 - 1.9,15/11/2023,Chromium,,56,mg/kg
BH05,1.8 - 1.9,15/11/2023,Copper,,22,mg/kg
BH05,1.8 - 1.9,15/11/2023,Lead,,26,mg/kg
BH05,1.8 - 1.9,15/11/2023,Mercury,<,0.1,mg/kg
BH05,1.8 - 1.9,15/11/2023,Nickel,,55,mg/kg
BH05,1.8 - 1.9,15/11/2023,Zinc,,344,mg/kg
BH05,5 - 5.1,15/11/2023,Arsenic,,12,mg/kg
BH05,5 - 5.1,15/11/2023,Cadmium,,2,mg/kg
BH05,5 - 5.1,15/11/2023,Chromium,,34,mg/kg
BH05,5 - 5.1,15/11/2023,Copper,,1460,mg/kg
BH05,5 - 5.1,15/11/2023,Lead,,557,mg/kg
BH05,5 - 5.1,15/11/2023,Mercury,<,0.1,mg/kg
BH05,5 - 5.1,15/11/2023,Nickel,,80,mg/kg
BH05,5 - 5.1,15/11/2023,Zinc,,30000,mg/kg
BH05,7.5 - 7.6,15/11/2023,Arsenic,<,5,mg/kg
BH05,7.5 - 7.6,15/11/2023,Cadmium,<,1,mg/kg
BH05,7.5 - 7.6,15/11/2023,Chromium,,52,mg/kg
BH05,7.5 - 7.6,15/11/2023,Copper,,43,mg/kg
BH05,7.5 - 7.6,15/11/2023,Lead,,68,mg/kg
```

BH05,7.5 - 7.6,15/11/2023,Mercury,<,0.1,mg/kg
BH05,7.5 - 7.6,15/11/2023,Nickel,,79,mg/kg
BH05,7.5 - 7.6,15/11/2023,Zinc,,1950,mg/kg

Chem Name,Action Level
Lead,300 mg/kg
Mercury,40 mg/kg
Nickel,400 mg/kg
Arsenic,100 mg/kg
Cadmium,20 mg/kg
Chromium (total),100 mg/kg
Copper,6000 mg/kg
Zinc,7400 mg/kg

You can do this in a single formula by:

1. Looking up the **Action Level** by *Chem Name*
2. Converting text like `"300 mg/kg"` → numeric
3. Adjusting results where Prefix = `<` (divide by 2)
4. Comparing the two

Assuming:

- Results table columns:
  - `Chem Name` = D
  - `Prefix` = E
  - `Result` = F
- Criteria table:
  - `Chem Name` = A
  - `Action Level` = B

✅ Formula

```
=IF(
   IF(E2="<",F2/2,F2) >
   VALUE(LEFT(XLOOKUP(D2,Criteria!A:A,Criteria!B:B),FIND(" ",XLOOKUP(D2,Criteria!A:A,Criteria!B:B
   "Exceeds",
   "OK"
)
```

🔧 What it does

- `XLOOKUP(D2,Criteria!A:A,Criteria!B:B)` → finds matching Action Level
- `LEFT(...,FIND(" ",...)-1)` → strips `" mg/kg"`
- `VALUE(...)` → converts to number
- `IF(E2="<",F2/2,F2)` → halves values with `<` prefix
- Final `IF` → compares result vs criterion

⚠️ Important note

Your criteria list has "**Chromium (total)**", but results use "**Chromium**".
This will cause a mismatch.

Fix options:

- Either change criteria to `"Chromium"`
- OR use a wildcard match version:

```
=IF(
   IF(E2="<",F2/2,F2) >
   VALUE(LEFT(XLOOKUP("*"&D2&"*",Criteria!A:A,Criteria!B:B,,2),FIND(" ",XLOOKUP("*"&D2&"*",Criter
   "Exceeds",
   "OK"
)
```

If you want, I can also give you a cleaner version using **LET()** (easier to read and faster), or convert this into a structured table formula.

which approach is more accurate

Short answer: **the exact match approach is more accurate**.

Here's why:

✅ Exact match (`XLOOKUP(D2, ...)`)

- Matches **only the intended chemical**
- No ambiguity or accidental matches
- Safer for regulatory or reporting work (like HIL comparisons)
- Prevents errors if you later add similar names (e.g. *Chromium VI*, *Chromium III*)

👉 This is the **recommended approach** for environmental / lab datasets.

⚠️ Wildcard match (`"*"&D2&"*"`)

- Matches **anything containing the text**
- Can return the **wrong criteria** if names are similar
  - e.g. "Chromium" could match:
    - Chromium (total)
    - Chromium VI
    - Chromium III
- Less reliable as your dataset grows

👉 Useful as a quick fix, but **not ideal for long-term accuracy**

🧠 Best practice (what I'd actually do)

Instead of relying on fuzzy matching:

✔️ Standardise names between tables:

- Change **"Chromium (total)"** → **"Chromium"**
  OR
- Add a helper column in criteria like:
  - `Match Name = "Chromium"`

Then use exact match.

🏁 Bottom line

- **Most accurate:** Exact match with clean, standardised names
- **Acceptable fallback:** Wildcard match (but riskier)

If you want, I can show you a version that **handles name mismatches safely without wildcards** (using a mapping table)—that's usually the most robust solution for this kind of dataset.

that sounds perfect

Great—that's the most robust way to handle this 👍

🧠 Best Practice: Use a Mapping Table

Instead of relying on messy name matching, create a small **mapping table** that links your *Results names* to *Criteria names*.

Example Mapping Table (new sheet: `Map`)

Result Name	Criteria Name
Chromium	Chromium (total)
Lead	Lead
Mercury	Mercury
Nickel	Nickel
Arsenic	Arsenic
Cadmium	Cadmium
Copper	Copper
Zinc	Zinc

✅ Final Formula (Robust + Accurate)

```
=LET(
    chem, D2,
    prefix, E2,
    result, F2,
    mappedChem, XLOOKUP(chem, Map!A:A, Map!B:B, chem),
    critText, XLOOKUP(mappedChem, Criteria!A:A, Criteria!B:B),
    critValue, VALUE(LEFT(critText, FIND(" ", critText)-1)),
    adjResult, IF(prefix="<", result/2, result),
    IF(adjResult > critValue, "Exceeds", "OK")
)
```

🔍 Why this is better

- ✅ **100% controlled matching** (no accidental hits)
- ✅ Handles naming differences cleanly
- ✅ Easy to maintain (just update the mapping table)
- ✅ Scales well if more chemicals are added later

⚠️ Extra tip (worth doing)

Wrap lookups in `IFERROR` to catch missing mappings:

excel

That way you'll immediately spot any chemicals that aren't mapped.

🏁 Summary

- Mapping table = **most accurate + safest**
- LET formula = **clean + efficient**
- Much better than wildcard matching for real-world datasets

If you want, I can also:

- Turn this into a **traffic light system (red/green)**
- Or build a **summary table of exceedances by borehole/depth**

how do i build a summary table of exceedances by borehole/depth

Nice—this is where your dataset becomes really useful.

You've already got the hard part (the comparison). Now we just summarise it.

✅ Option 1 (Best): Pivot Table

Step 1: Add a helper column

Use your existing formula to return `"Exceeds"` or `"OK"` (let's say it's in column **H**, called `Status`).

Step 2: Insert Pivot Table

1. Select your results table
2. Go to **Insert → PivotTable**
3. Place it in a new sheet

Step 3: Build the Pivot

Drag fields like this:

- **Rows**
  - `Location Code`
  - `Depth`
- **Columns**
  - `Status`
- **Values**
  - `Status` → set to **Count**

Result:

Location	Depth	Exceeds	OK
BH01	3–3.1	6	2
BH01	4.8–4.9	4	4
...	...	...	...

🎯 Optional improvements

- Filter to show only `"Exceeds"`
- Add `Chem Name` as a filter or slicer
- Add conditional formatting (red for exceedances)

✅ Option 2: Formula-Based Summary (Dynamic)

If you prefer formulas (no Pivot):

Step 1: Get unique combinations

```excel
```

(Assumes `Location Code` in A, `Depth` in B)

Step 2: Count exceedances

Next to it:

```excel
```

Where:

- J2 = Location Code
- K2 = Depth
- H:H = Status column

Step 3: Count OKs (optional)

```excel
```

🔥 Option 3 (Best of both): One Formula Summary

If you're on Excel 365:

```excel
=LET(
    loc, A2:A100,
    depth, B2:B100,
    status, H2:H100,
    combos, UNIQUE(HSTACK(loc, depth)),
    exceeds, BYROW(combos, LAMBDA(r,
        COUNTIFS(loc, INDEX(r,1), depth, INDEX(r,2), status, "Exceeds")
    )),
    HSTACK(combos, exceeds)
)
```

🧠 What I recommend

- **Quick + flexible → Pivot Table**
- **Automated reporting → Formula version**
- **Large datasets → Pivot + slicers**

🚀 Bonus idea (very useful in reporting)

Instead of counts, flag if *any exceedance exists*:

```excel
=IF(COUNTIFS(A:A,J2,B:B,K2,H:H,"Exceeds")>0,"Fail","Pass")
```

This gives you a clean compliance summary per borehole/depth.

If you want, I can take your exact dataset and show what the finished summary should look like (including which boreholes fail).